

Question 3: Should knowledge in an area of knowledge be pursued for its own sake rather than for its potential application? Discuss with reference to mathematics and one other area of knowledge.

1535 words

Questions of motivation and truth come to light when discussing the development and acquisition of knowledge. Knowledge pursued for its own sake is the discovery of new information, by knowledge seekers' with the intent to satisfy their curiosity and desire to expand their knowledge, that has no utilitarian value. In contrast, knowledge that is pursued for its potential application, such as novel inventions, is the acquisition of new information by knowledge seekers with the motivation of using it in the real world, such as to solve a problem or to find a more efficient way of completing tasks. Though critics may classify the pursuit of knowledge for its own sake as aimless exploration with no foreseeable use, such knowledge can still be appreciated by those who admire what can be obtained instead of what must be obtained. This essay aims to evaluate the value in the pursuit of knowledge for its own sake versus that of knowledge pursued for potential application by exploring their implications on society, in the areas of knowledge (AOK) of mathematics and the natural sciences.

Debatably, knowledge should be pursued for its own sake as it is likely to improve the understanding of an AOK by introducing new ways of approaching a problem. The actions of Srinivasa Ramanujan would be one such example. Ramanujan, an Indian mathematician, had become obsessed with deducing mathematical proofs of theorems that he read about, within the field of pure mathematics (The Editors of Encyclopaedia Britannica; Neville 292). Rejecting the route of higher education, he spent five years pursuing the knowledge of proofs behind equations in pure maths,

leading to his discovery of mathematical knowledge in number theory and infinite series (The Editors of Encyclopaedia Britannica). Such knowledge had left professional mathematicians in awe due to its novelty (Neville 293; Kanigel 168). Ramanujan's desire and determination to pursue mathematical knowledge allowed him to exceed the imaginations of professional mathematicians, who greatly appreciated and admired his findings. Furthermore, he was able to provide mathematicians with a rare opportunity of a different perspective from outside mainstream intellectual thinking, availing to them an opportunity to better understand pure maths.

However, it could be argued that the knowledge acquired by Ramanujan has no application in the real world. Also, society outside mathematical circles may find it difficult to appreciate his pursuit of knowledge and would not benefit as much as a professional mathematician, given that a steep learning curve exists for the layman to fully grasp the knowledge Ramanujan acquired. Nevertheless, his pursuit of the knowledge for the knowledge's own sake has revealed the self-evident value of his strategy: introducing unique perspectives that had not been seen before. Hence, knowledge should be pursued for its own sake, especially within a formalistic view of pure mathematics.

Plausibly, knowledge should be pursued for its own sake in the natural sciences as it enables other concepts within the AOK to be expressed in different ways. This has been demonstrated in string theory, a concept in particle physics, which aims to explain the fabric of the universe (Cvetič et al.). Using string theory, various physics theories have been consolidated under that single framework, thereby providing different perspectives to the said concepts (Greene; Cvetič et al.). As string theory

has never been proven through empirical evidence, physicists and theorists are still pursuing knowledge within the theory, to this day (Greene).

Pursued for its own sake, the knowledge within string theory in physics is seen as valuable, given that knowledge seekers still choose to pursue it even though there is no application of the theory in the physical world. However, string theory could serve as an extendable and understandable framework that serves to strengthen existing knowledge in physics by enabling different concepts to be linked and built on top of one another.

Arguably, knowledge should also be pursued for its potential application as it can be used to solve a problem while using the time of knowledge seekers efficiently. Knowledge in mathematics, applied in computer science, has been used to secure sensitive digital information through encryption, where encryption algorithms rely on mathematical principles. This is demonstrated in the development of the Rijndael, an encryption algorithm which utilised the mathematics theorem of finite fields in order to better obscure the original meaning of information that had been encrypted (Daemen 9). Rijndael's development was imperative as its preceding encryption algorithm had become antiquated and was increasingly susceptible to cyber attacks (Smid 2). Since its introduction, Rijndael has been widely adopted in society, providing security to government and private firms' data (Smid 14-5; Nord Security). With the use of applied mathematics, a solution to the growing problem of a weakening encryption algorithm was developed, with individuals uninvolved in the development of Rijndael benefiting from enhanced security standards Rijndael introduced. Moreover, the fact that the developers of Rijndael had a clear goal of replacing an antiquated encryption algorithm allowed them to pursue and apply the

knowledge of finite fields in the development of Rijndael, which provided society with powerful protection for their data in the digital age.

However, it is entirely probable that knowledge of other well-designed encryption algorithms were upstaged by the success of Rijndael, where many early adopters of Rijndael never considered other kinds of encryption algorithms. As such, it is possible that the necessity to develop more knowledge akin to Rijndael was mostly extinguished, depriving other knowledge seekers of the opportunity of pursuing knowledge for potential application just because it would be an act of reinventing existing technology.

Still, the purpose of pursuing knowledge for potential application is to utilise knowledge seekers' time in the most efficient way by solving targeted problems. This way, other knowledge seekers would be able to focus their attention on other pressing matters, making full use of available knowledge seekers within an AOK. Therefore, knowledge in mathematics should be pursued for its potential application, given its applicability to improving systems through optimal use of resources.

It is also possible that knowledge in the natural sciences, pursued for its potential application, can be acquired to provide logical explanations for enigmatic circumstances, such as in aviation accidents. One such example would be the investigation of the fatal mid-air disintegrations of three de Havilland Comet airliners, involving the pursuit of knowledge related to pressure within the field of physics. The investigation found that the repeated cycles of pressurising the aircraft with air at high altitude and the subsequent depressurisation of the aircraft after it had landed created stress on the fuselage of the Comet, especially at the sharp corners of its square-shaped windows (Johnston; Glancey; Goyer; Royal Air Force Museum). The

pursuit of knowledge in physics for its potential application in explaining the mysterious loss of the aircrafts allowed knowledge seekers, such as the investigators of the accidents, to obtain a precise explanation and series of unfortunate circumstances that had occurred. Also, the acquisition of said knowledge could have provided closure to those affected by the loss of life from the accidents. In this context, the pursuit of knowledge was a crucial step in progressing society intellectually and emotionally.

However, pursuing knowledge for its potential application in this way poses a high risk to society. As seen in the case of the three Comet disintegrations, it was only after the tragic loss of life and a clear indication of a design flaw which prompted the pursuit of knowledge. By acquiring knowledge for potential application in society, said knowledge should not pose a safety risk, especially if human life is dependent on the reliability and accuracy of said knowledge. However, no system may always be perfect in preventing the unanticipated from happening. Hence, the pursuit of knowledge for its potential application in obtaining a cause behind unexplained events should be conducted by knowledge seekers given the importance of identifying mistakes as a prerequisite to learn from them.

The implications of knowledge pursued for its potential application include the efficient use of scarce resources to accomplish a specific task as well as being the grounds for which humanity can improve by utilising it as a means to adapt in response to changes that may occur without an immediate explanation. On the other hand, knowledge pursued for its own sake implies that knowledge seekers have the drive to acquire knowledge despite opposition or the lack of support from the fraternity before acknowledgement. Furthermore, this allows knowledge to be

acquired in novel and intriguing ways, which expands the understanding of other knowledge seekers within a niche part of an AOK.

In conclusion, knowledge should either be pursued for its own sake or for potential application, where neither option presents an overall advantage over the other. However, in a given situation, AOK or a group of knowledge seekers may prefer or necessitate the pursuit of knowledge for its own sake instead of pursuing it for potential application, or vice versa. For example, physical events that trigger the pursuit for knowledge have been shown to involve the knowledge's potential application in the physical event, such as when Rijndael was created in response to an increasingly lacking encryption algorithm. Knowledge seekers should therefore evaluate their circumstance and personal desires to decide on pursuing knowledge for its own sake or for potential application.

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